

In-Memory Caching

In-memory caching stores data in RAM instead of disk, providing sub-millisecond access times. It sits between your application and database to reduce load and improve performance.

Typical latency comparison:

- RAM (cache): 0.1-1 ms
- SSD database: 1-10 ms
- HDD database: 10-100 ms
- Network database call: 10-50+ ms

Feature	Redis	Memcached
Data types	Rich data types (strings, lists, sets, sorted sets, hashes, bitmaps, HyperLogLog)	Pure key-value storage (Strings only)
Persistence	Optional persistence (RDB snapshots, AOF logs)	No Persistence
Replication	Built-in replication and clustering	No replication built-in
Performance	Very fast	Extremely fast for simple keys
Memory efficiency	Lower	Higher
Complexity	More	Simple
Multi-threading	Multi-threaded (better CPU utilization)	Single-threaded
Eviction	Multiple eviction policies	Simple LRU eviction
Best For	• Complex caching scenarios • When you need data structures • Rate limiting, leaderboards, counters • Session management with structure • Real-time analytics • Message queues	• Simple caching scenarios • When you need pure speed • Storing session data • Cache-aside pattern